

Structure & Bonding of the Period 3 Elements (Cambridge (CIE) A Level Chemistry): Revision Note

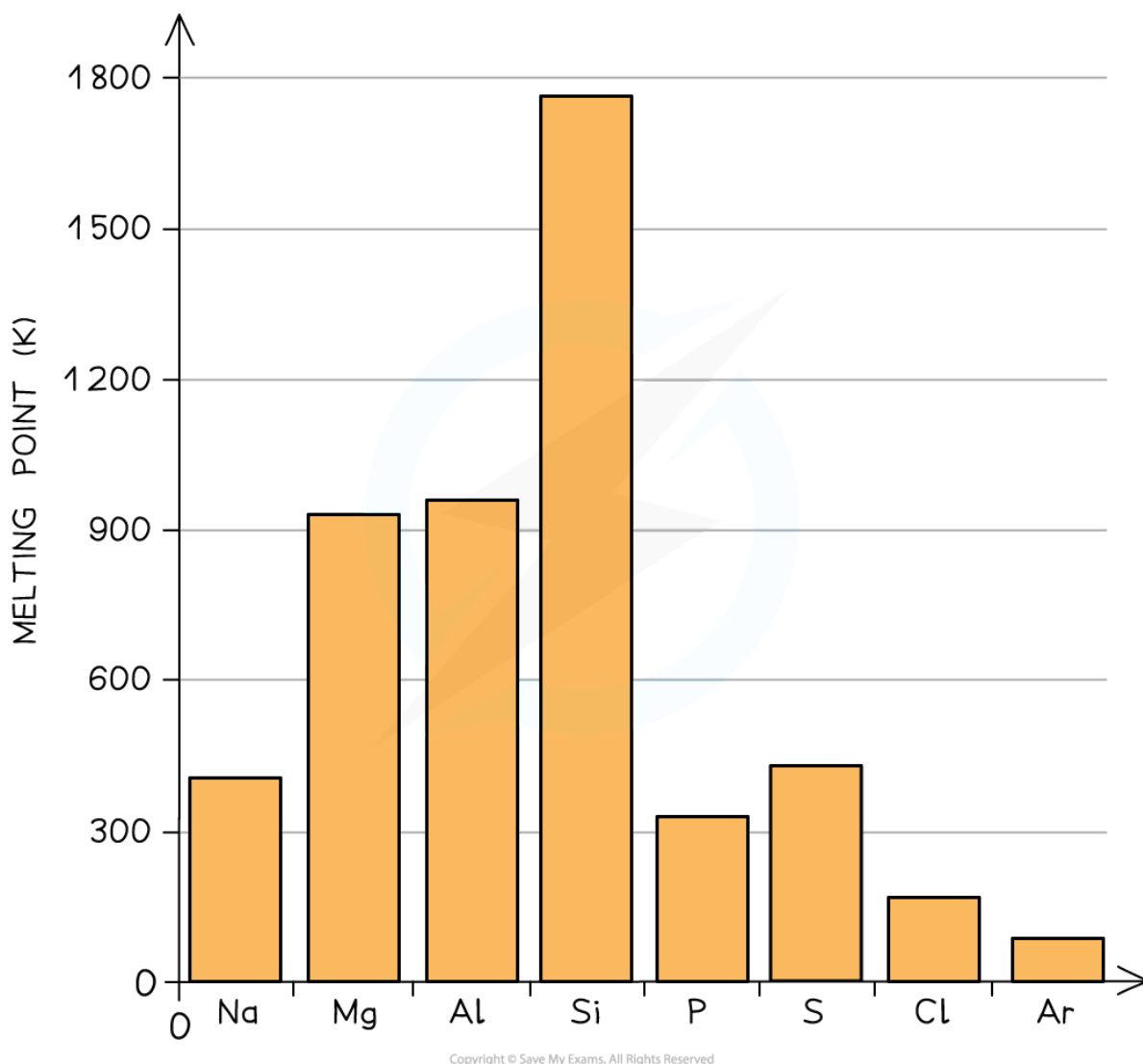
Explaining Physical Properties of the Period 3 Elements

Melting point

Melting points of the elements across Period 3 table

Period 3 element	Na	Mg	Al	Si	P	S	Cl	Ar
Melting point (K)	371	923	932	1638	317	392	172	84

Graph of melting points across Period 3



There is a general increase in melting point from Na to Si, followed by a sharp drop to the lower melting points of P to Ar

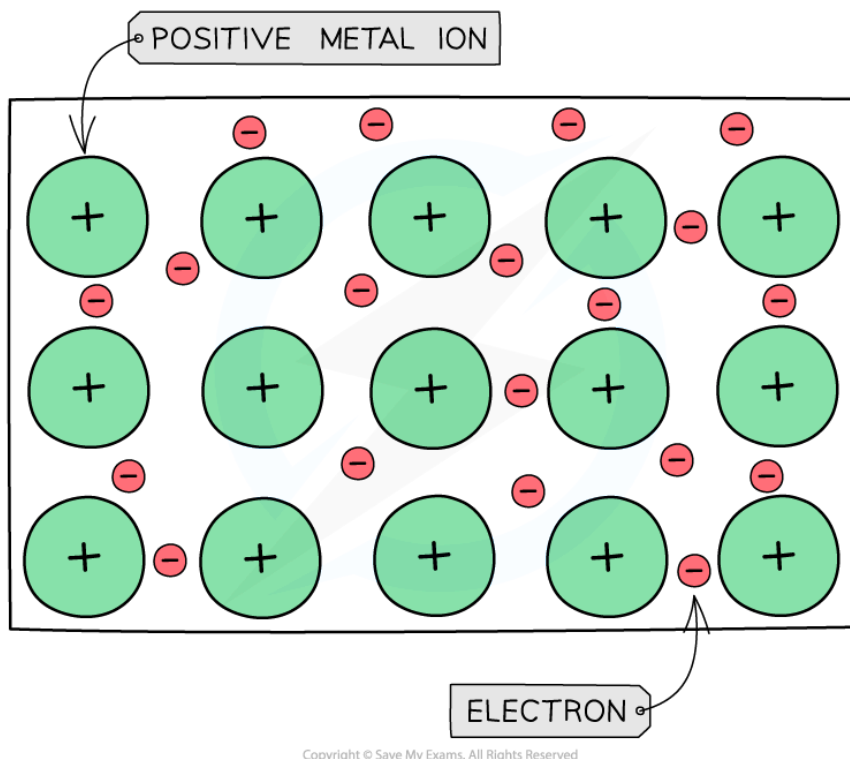
- The above trends can be explained by looking at the bonding and structure of the elements

Bonding and structure of the Period 3 elements table

Period 3 element	Na	Mg	Al	Si	P	S	Cl
Bonding	Metallic	Metallic	Metallic	Covalent	Covalent	Covalent	Covale
Structure	Giant metallic	Giant metallic	Giant metallic	Giant molecular	Simple molecular	Simple molecular	Simpl molecu

- The table shows that **Na**, **Mg** and **Al** are metallic elements which form positive ions arranged in a **giant lattice** in which the ions are held together by a 'sea' of delocalised electrons around them

The structure of metals



Metal cations form a giant lattice held together by electrons that can freely move around

Na, Mg and Al

- The electrons in the 'sea' of delocalised electrons are those from the **valence shell** of the atoms
- Na** will donate one electron into the 'sea' of delocalised electrons, **Mg** will donate two and **Al** three electrons
- As a result of this, the metallic bonding in **Al** is stronger than in **Na**
- This is because the electrostatic forces between a **3+ ion** and the larger number of negatively charged delocalised electrons is much larger compared to a **1+ ion** and the smaller number of delocalised electrons in Na
- Due to this, the **melting points increase** going from **Na** to **Al**

Si

- Si** has the highest melting point due to its giant molecular structure in which each Si atom is held to its neighbouring Si atoms by **strong covalent bonds**

P, S, Cl and Ar

- **P, S, Cl** and **Ar** are non-metallic elements and exist as **simple molecules** (P_4 , S_8 , Cl_2 and Ar as a single atom)
- The **covalent bonds within** the molecules are strong, however, **between** the molecules, there are only weak **instantaneous dipole-induced dipole forces**
- It doesn't take much energy to break these **intermolecular** forces
- Therefore, the melting points decrease going from **P** to **Ar** (note that the melting point of S is higher than that of P as sulphur exists as larger S_8 molecules compared to the smaller P_4 molecule)

Electrical conductivity

- The **electrical conductivity decreases** going across the Period 3 elements

Trends in electrical conductivity across Period 3 table

Period 3 element	Na	Mg	Al	Si	P	S	Cl	Ar
Electrical conductivity ($S\ m^{-1}$)	0.218	0.224	0.382	2×10^{-10}	10^{-17}	10^{-23}	-	-

- Going from **Na** to **Al**, there is an increase in the number of valence electrons that are donated to the 'sea' of delocalised electrons
- Because of this, in **Al** there are more electrons available to move around through the structure when it conducts electricity, making **Al** a better electrical conductor than **Na**
- Due to the **giant molecular structure** of **Si**, there are no delocalised electrons that can freely move around within the structure
- **Si** is therefore not a good electrical conductor and is classified as a **semimetal (metalloid)**
- The lack of delocalised electrons is also why **P** and **S** cannot conduct electricity



Examiner Tips and Tricks

- **Intermolecular forces** are forces **between** molecules
- **Intramolecular forces** are forces **within** a molecule